Weizheng Wang

weizheng.wang.cs@gmail.com wz-wang.com github.com/4everWZ

Research interests

I develop controlled evaluations for language and multimodal models, separating semantic decision errors from structural and parsing failures. My research interests are reliable agent behavior in sequential decision-making, outcome-based evaluation under noisy inputs, and efficient deployment. I aim to extend my offline evaluation and edge-inference work to agent systems that must act and recover reliably.

Education

Adelaide University, Australia 2025–2026

Master of Artificial Intelligence and Machine Learning | Conferred October 2026

Completed the two-year master's programme within one year.
Program GPA: 6.75/7.0 (original scale). Research Project Part A: 90; Part B: 87.

North China Institute of Science and Technology, China 2021–2025

Bachelor of Engineering in Computer Science and Technology | Graduated 2025

GPA: 4.02/5.0 (original scale).

Selected research and engineering experience

Zayed University | Research Assistant, remote Feb.–Aug. 2026

MulRobBench, manuscript submitted | Supervisor: Dr Belal S. Alsinglawi

- Co-defined protocol-conditioned UAV tasks and built data-generation, inference and evaluation pipelines for 3,024 offline samples and 17 models under visual degradation and noisy instructions.
- Separated semantic decision quality, unsafe actions, structural compliance and parsing failures. Semantic, structural and action metrics produced different model rankings.

Institute of Software, Chinese Academy of Sciences Feb.–Jun. 2025

Research Intern, remote | MindNLP / MindSpore model integration

- Implemented and fine-tuned Autoformer for time-series modeling, BEiT for vision and ALBERT for language in MindNLP/MindSpore.
- Addressed training stability and reproducibility during framework integration; contributed code reviews, issue handling and documentation.

Beijing Elite Intelligence Company & NCIST Jan.–Jul. 2024

Research Intern | UAV mine inspection and edge perception

- Developed UAV-video perception for mine inspection: standing-water counting, belt-deviation detection and truck monitoring from aerial video using YOLOv8 detection and DeepSORT tracking.
- Optimized PyTorch detectors and exported models through ONNX to connect inspection functions with the UAV edge-inference pipeline.

Selected implementation projects

DEIM Jetson Acceleration | Converted detector models through ONNX/TensorRT and checked FP16/INT8 correctness, latency and power. Selective INT8 was slower than FP16 in particular profiled cases, showing why precision choices require device-specific measurement.

Bachelor's thesis | May 2025

Research and implementation of substation alarm detection system based on multi-modal image and large language model

Supervisor: Hongjun Ju | Video-based Intelligent Alert System

Integrated multi-stream video monitoring, open-vocabulary detection, alerts and multimodal question answering in a visual monitoring application.

Selected submitted manuscripts 2026

When Retrieval Is Not Binding: A Diagnostic Benchmark for Visual Episodic Memory

W. Wang, J. Wu et al. | Equal contribution. *Submitted to IEEE ICASSP 2027.*

Designed matched memory-off, wrong-entity and shuffled-binding interventions while holding queries, candidates and features fixed. The tests distinguish retrieved-content benefit from sensitivity to episodic binding.

FHS-DEIM: Frequency-Aware Hybrid Encoding with Spiral Mamba for Infrared UAV Search and Rescue

Submitted to Journal of King Saud University Computer and Information Sciences (2026).

Master's research project

Knowledge Graph Contraction and Visual Exploration for Australian Skills and Occupations Open Data

Master's Research Project Parts A/B | Supervisor: Dr Gun Lee | July 2026

Built a local data-to-visualization workflow for 1,236 occupations and 19 industries, linking graph construction and Louvain community-based contraction to interactive search, source inspection and graph-grounded explanations. In the reported snapshot, nodes decreased from 1,255 to 609 while edges increased from 2,743 to 3,628: a smaller node-level overview with additional summary and similarity links.

Selected publications

A Lightweight Thermal Denoising and Occlusion-Robust Infrared Detection Model for Substation Equipment. J. Wu, **W. Wang**, Y. Niu, X. Shi, L. Tian. IEEE ICIP, 2026. PO-YOLO: 83.5% mAP50 / 56.9% mAP50:95 on ISED-1K. Code.

A lightweight insulator defect detection algorithm based on drone images for power line inspection. J. Wu, **W. Wang**, W. Du et al. (W. Wang and W. Du: equal contribution). Engineering Research Express 7, 045355, 2025. DOI: 10.1088/2631-8695/ae19ce.

Real-time monitoring of trucks used in open pit based on aerial video of UAV. W. Wang, J. Wu, Y. Li. IEEE RAIIC, pp. 130–138, 2024. DOI: 10.1109/RAIIC61787.2024.10671199.

Research on industrial meter readings based on improved YOLOv8. J. Wu, **W. Wang**, L. Yang, X. Ni, L. Tian. Proc. SPIE 13181, 131815G, 2024. DOI: 10.1117/12.3031129.

Scholarship, leadership and honours

2025: Global Citizens International Scholarship (highest tier), University of Adelaide.
2024: Project lead, National Undergraduate Innovation Training Program (202411104021): UAV perception, automated inspection and edge inference.
2023: Grand Award, National 3D Digitization Innovation Competition; Second Prize, Challenge Cup (Science and Technology Invention and Creation Track); First Prize, National College Students' Data Analysis Competition.

Technical skills

Development: Python, C++, Git, Linux/WSL, Docker, CMake.
Learning and vision: PyTorch, MindSpore, Transformers, OpenCV, DeepSORT.
Deployment and streaming: ONNX, TensorRT, Jetson, ZLMediaKit/FFmpeg.
Languages: Chinese, English.

Date: 1 November 2026